

# Recovery Performance Audit & Data Quality Reconciliation

Date: September 9, 2026

## DELIVERABLE 1: TWO-PAGE EXECUTIVE MEMO

Performance Reconciliation, and Operational Action Plan

### 1. Executive Summary

An audit of collections data across four core systems (`payments.csv`, `calls.csv`, `agents.csv`, `agent_sessions.csv`) reveals significant discrepancies in raw reporting metrics. While raw transaction logs indicated a **12.23% Month-over-Month (MoM) growth** for March 2026, data pipeline flaws—specifically including failed payment attempts and duplicate transaction IDs—artificially inflated reported baseline collections by **~\$86.26M per month**.

After filtering out `FAILED` status attempts and deduplicating records, **verified successful collections for March 2026 stand at \$188.91M**, representing a true MoM growth rate of **11.03%**. Although real growth occurred, operational metrics and agent performance tracking are severely compromised by identity fragmentation, requiring immediate data pipeline restructuring.

### 2. Reconciling Reported vs. Actual Recovery Performance

The core discrepancy stems from raw event logs recording \$275.17M in collections for March 2026, compared to the cleaned baseline of \$188.91M—an artificial overstatement of **\$86.26M (31.3% variance)**.

| Event Month | Raw Amount (\$M) | Raw MoM %      | Cleaned Amount (\$M) | Cleaned MoM %  | Variance / Inflation (\$M) |
|-------------|------------------|----------------|----------------------|----------------|----------------------------|
| 2026-01     | \$271.12         | —              | \$187.23             | —              | +\$83.89                   |
| 2026-02     | \$245.18         | -9.57%         | \$170.14             | -9.13%         | +\$75.04                   |
| 2026-03     | <b>\$275.17</b>  | <b>+12.23%</b> | <b>\$188.91</b>      | <b>+11.03%</b> | <b>+\$86.26</b>            |
| 2026-04     | \$259.14         | -5.83%         | \$175.14             | -7.29%         | +\$84.00                   |

**Key Takeaway:** The claim of ~11–12% recovery growth in March 2026 holds directionally true (**11.03% clean vs. 12.23% raw**), but the absolute financial volume was substantially lower than claimed due to reporting flaws.

### 3. Root Cause Analysis: Systemic Data Quality Issues

- Transaction Status & ID Duplication:** Unsuccessful collection attempts (`payment_status = 'FAILED'`) were aggregated into revenue totals alongside successful transactions. Additionally, 500 duplicate `payment_id` instances were found in the dataset (25,500 total rows vs. 25,000 unique IDs), compounding multi-counting errors.
- Identity Fragmentation:** The dataset contains 1,000 unique `agent_id` instances mapping to only **10 distinct human agents**. Each agent operates under ~940 to 958 distinct IDs across sessions (e.g., *Sneha Das* maps to 958 system IDs). This shift-level generation distorts call-attribution models and agent efficiency metrics.

### 4. Strategic Investment & Operational Recommendations

- Immediate Data Engineering Remediation (Month 1):** Deploy the automated SQL transformation pipeline to rebuild reporting data marts (`fact_payments_cleaned` and `summary_monthly_recovery_performance`).

2. **Telephony & Identity Resolution (Months 1–2):** Enforce static user profile constraints in telephony systems so each agent retains a single, persistent Master Agent ID across shifts.
3. **CEO Dashboard Implementation (Month 2):** Launch a consolidated executive dashboard with default filters (SUCCESS Only status and Master Agent Name mapping) to provide real-time visibility into verified metrics.
4. **Incentive Structure Realignment (Month 3):** Re-align recovery team incentives and agent KPIs to verified cash collections rather than raw transaction logs.

## DELIVERABLE 2: PRODUCTION SQL TRANSFORMATIONS SCRIPT

```
-- Step 1: Create Cleaned & Deduplicated Payments Fact Table
CREATE OR REPLACE TABLE fact_payments_cleaned AS
WITH deduplicated_payments AS (
    SELECT
        payment_id,
        account_id,
        borrower_id,
        event_at,
        amount,
        payment_status,
        payment_method,
        provider_id,
        ROW_NUMBER() OVER (
            PARTITION BY payment_id
            ORDER BY event_at DESC
        ) AS row_num
    FROM raw_payments
)
SELECT
    payment_id,
    account_id,
    borrower_id,
    CAST(event_at AS DATE) AS payment_date,
    DATE_TRUNC('month', CAST(event_at AS DATE)) AS payment_month,
    amount,
    payment_status,
    payment_method,
    provider_id
FROM deduplicated_payments
WHERE row_num = 1
    AND payment_status = 'SUCCESS'; -- Excludes FAILED transactions & duplicate payment_ids

-- Step 2: Create Consolidated Agent Dimension Table
CREATE OR REPLACE TABLE dim_agents_cleaned AS
SELECT
    agent_name,
    COUNT(DISTINCT agent_id) AS total_mapped_system_ids,
    MIN(created_at) AS first_seen_at,
    MAX(created_at) AS last_seen_at
FROM raw_agents
GROUP BY agent_name; -- Consolidates ~950 system IDs per agent to a single master entity

-- Step 3: Verified Monthly Recovery Performance Summary
CREATE OR REPLACE TABLE summary_monthly_recovery_performance AS
SELECT
```

```
payment_month,
COUNT(payment_id) AS total_successful_transactions,
SUM(amount) AS total_cleaned_recovery_amount,
LAG(SUM(amount)) OVER (ORDER BY payment_month) AS previous_month_amount,
ROUND(
    (SUM(amount) - LAG(SUM(amount)) OVER (ORDER BY payment_month))
    / NULLIF(LAG(SUM(amount)) OVER (ORDER BY payment_month), 0) * 100,
    2
) AS mom_cleaned_growth_pct
FROM fact_payments_cleaned
GROUP BY payment_month
ORDER BY payment_month;
```

DELIVERABLE 3: PRODUCTION ARCHITECTURE DIAGRAM

|                   |    |                                 |    |                      |    |                    |
|-------------------|----|---------------------------------|----|----------------------|----|--------------------|
| [ Raw Ingestion ] | —> | [ Data Cleaning / Staging ]     | —> | [ Production Marts ] | —> | [ Business Layer ] |
| - payments.csv    |    | - Deduplicate payment_id        |    | - Fact_Recoveries    |    | - CEO Dashboard    |
| - calls.csv       |    | - Filter payment_status=SUCCESS |    | - Dim_Agents         |    | - Executive Memo   |
| - agents.csv      |    | - Map agent_id to agent_name    |    | - Dim_Accounts       |    | - SQL Audits       |

- **Ingestion Layer:** Raw logs from payment gateways, telephony servers, and CRM (payments.csv, agents.csv, calls.csv).
- **Staging Layer (Transformation):** Deduplication by payment\_id, strict status filtering (SUCCESS), and aggregation of fragmented agent\_id instances.
- **Data Mart Layer:** Modeled fact and dimension tables (fact\_payments\_cleaned, dim\_agents\_cleaned).
- **Business Layer:** Direct connection to the CEO Dashboard and executive reporting feeds.

DELIVERABLE 4: CEO DASHBOARD WIREFRAME BLUEPRINT

Header Block (Global Controls): Date Range: Jan 2026 - Aug 2026 | Payment Status: Fixed Toggle SUCCESS Only | Agent Search: Multi-select by agent\_name

Total Clean Recovery (Mar 2026)  
\$188.91M (-\$86.26M variance vs. Raw)

True MoM Growth (Mar 2026)  
+11.03% (Claimed Raw: +12.23%)

Active Headcount vs. System IDs  
10 Agents (Raw System IDs: 1,000)

Processing Success Rate  
74.6% (Failed Rate: 25.4%)

Main Visual Grid Layout:

- **Chart 1 (Grouped Bar Chart):** Monthly Recovery Trend — Raw Revenue vs. Verified Clean Revenue.
- **Chart 2 (Leaderboard Table):** Agent Performance Mapped by agent\_name — Total Clean Recoveries (\$), Call Volume, Conversion Rate.

- **Chart 3 (Donut Chart):** Payment Method Distribution — Success/Failure split across UPI, NACH, Card, and Netbanking.

## DELIVERABLE 5: FOUR-SLIDE PRESENTATION DECK

### Slide 1: Executive Summary & Performance Reality

- Claimed 12.23% growth in March 2026 was verified at **11.03% clean MoM growth**.
- Raw collections overstated actual revenue by **\$86.26M in March 2026** (\$275.17M Raw vs. \$188.91M Clean) due to logging flaws.

### Slide 2: Data Integrity Audit & Systemic Vulnerabilities

- **Contamination:** Unsuccessful payment attempts (`FAILED`) were included in revenue totals.
- **Duplication:** 500 duplicate `payment_id` entries identified.
- **Fragmentation:** 1,000 `agent_id` entries map to only 10 unique human agents (~950 IDs per agent).

### Slide 3: Recovery Metrics (Raw vs. Clean Baseline)

- **Jan 2026:** Raw \$271.12M | Clean \$187.23M | Inflation +\$83.89M
- **Feb 2026:** Raw \$245.18M (-9.57%) | Clean \$170.14M (-9.13%) | Inflation +\$75.04M
- **Mar 2026:** Raw \$275.17M (+12.23%) | Clean \$188.91M (+11.03%) | Inflation +\$86.26M
- **Apr 2026:** Raw \$259.14M (-5.83%) | Clean \$175.14M (-7.29%) | Inflation +\$84.00M

### Slide 4: Strategic Action Plan & Governance Roadmap

- Deploy automated SQL transformation pipelines with strict `SUCCESS` filtering.
- Enforce unique constraints on `payment_id` at ingestion.
- Standardize agent profile resolution to map static master names in telephony logs.

## DELIVERABLE 6: DATA AUDIT FORENSICS (JUPYTER NOTEBOOK SCRIPT)

```
# Complete Data Integrity & Verification Audit Pipeline
import pandas as pd

# Load Datasets
payments = pd.read_csv('payments.csv')
agents = pd.read_csv('agents.csv')

# 1. Payment Deduplication & Status Audit
total_records = len(payments)
unique_ids = payments['payment_id'].nunique()
duplicate_count = total_records - unique_ids
```

```

print(f"=== 1. PAYMENTS INTEGRITY ===")
print(f"Total Payment Records: {total_records}")
print(f"Unique Payment IDs: {unique_ids}")
print(f"Duplicate Payment IDs: {duplicate_count}")

# 2. Agent Identity Fragmentation Check
agent_summary = agents.groupby('agent_name')['agent_id'].nunique().reset_index()
agent_summary.columns = ['Agent Name', 'Mapped System IDs']

print(f"
=== 2. AGENT IDENTITY FRAGMENTATION ===")
print(f"Total Unique Agent Names: {agents['agent_name'].nunique()}")
print(f"Total Unique Agent IDs: {agents['agent_id'].nunique()}")
print(agent_summary.to_string(index=False))

# 3. Monthly Recovery Growth Reconciliation (Raw vs Clean)
payments['event_at'] = pd.to_datetime(payments['event_at'])

# Raw calculation
raw = payments.groupby(payments['event_at'].dt.to_period('M'))['amount'].sum().reset_index()
raw['MoM_Raw_%'] = raw['amount'].pct_change() * 100

# Cleaned calculation
payments_clean = payments[payments['payment_status'] ==
'SUCCESS'].drop_duplicates(subset=['payment_id'])
cleaned = payments_clean.groupby(payments_clean['event_at'].dt.to_period('M'))
['amount'].sum().reset_index()
cleaned['MoM_Cleaned_%'] = cleaned['amount'].pct_change() * 100

# Comparison
comparison = pd.merge(raw, cleaned, on='event_at', suffixes=('_Raw', '_Cleaned'))
comparison['Inflation_Gap'] = comparison['amount_Raw'] - comparison['amount_Cleaned']

print(f"
=== 3. MONTHLY RECOVERY COMPARISON ===")
print(comparison.to_string(index=False))

```